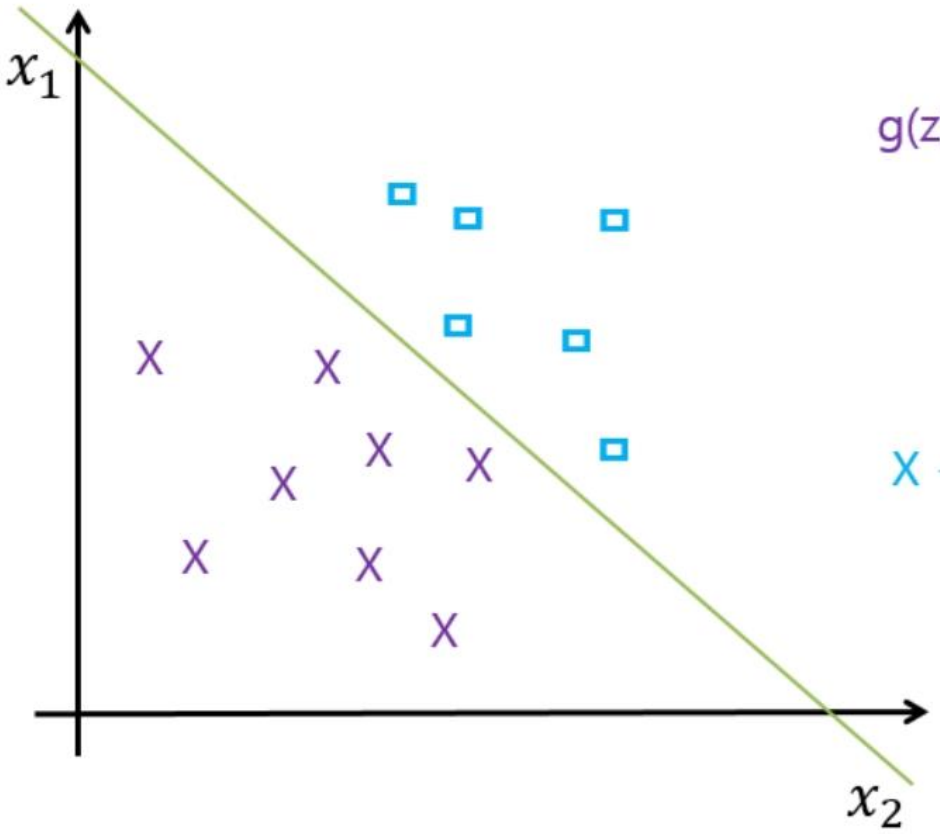
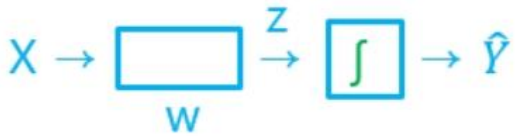
softmax

LOGISTIC REGRESSION

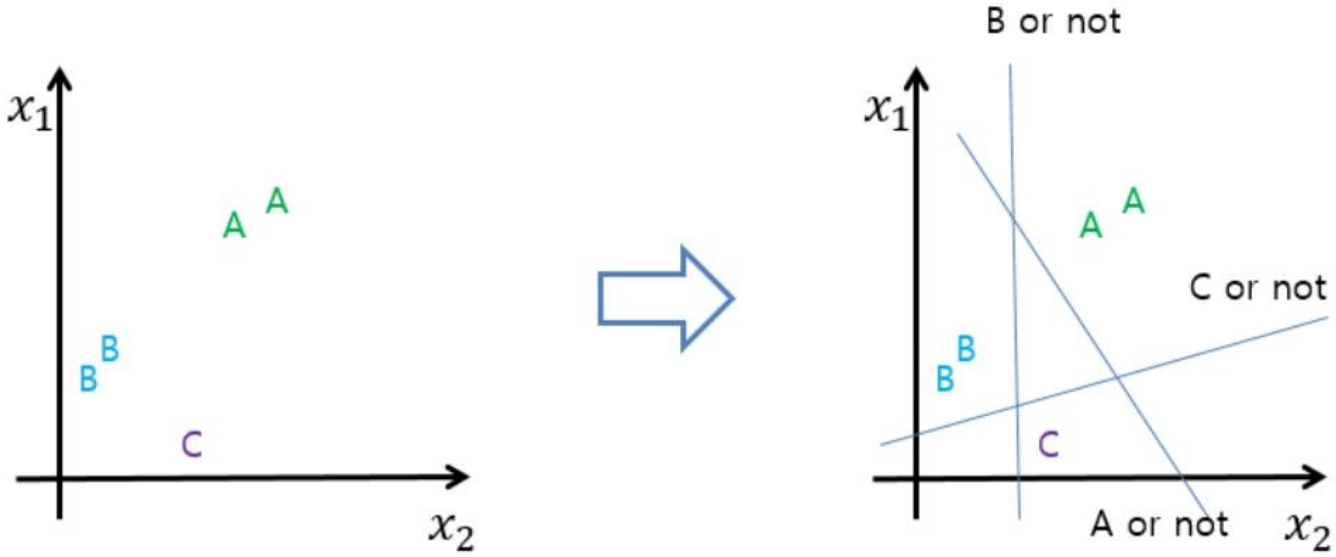


$$g(z) = \frac{1}{1+e^{-z}}$$

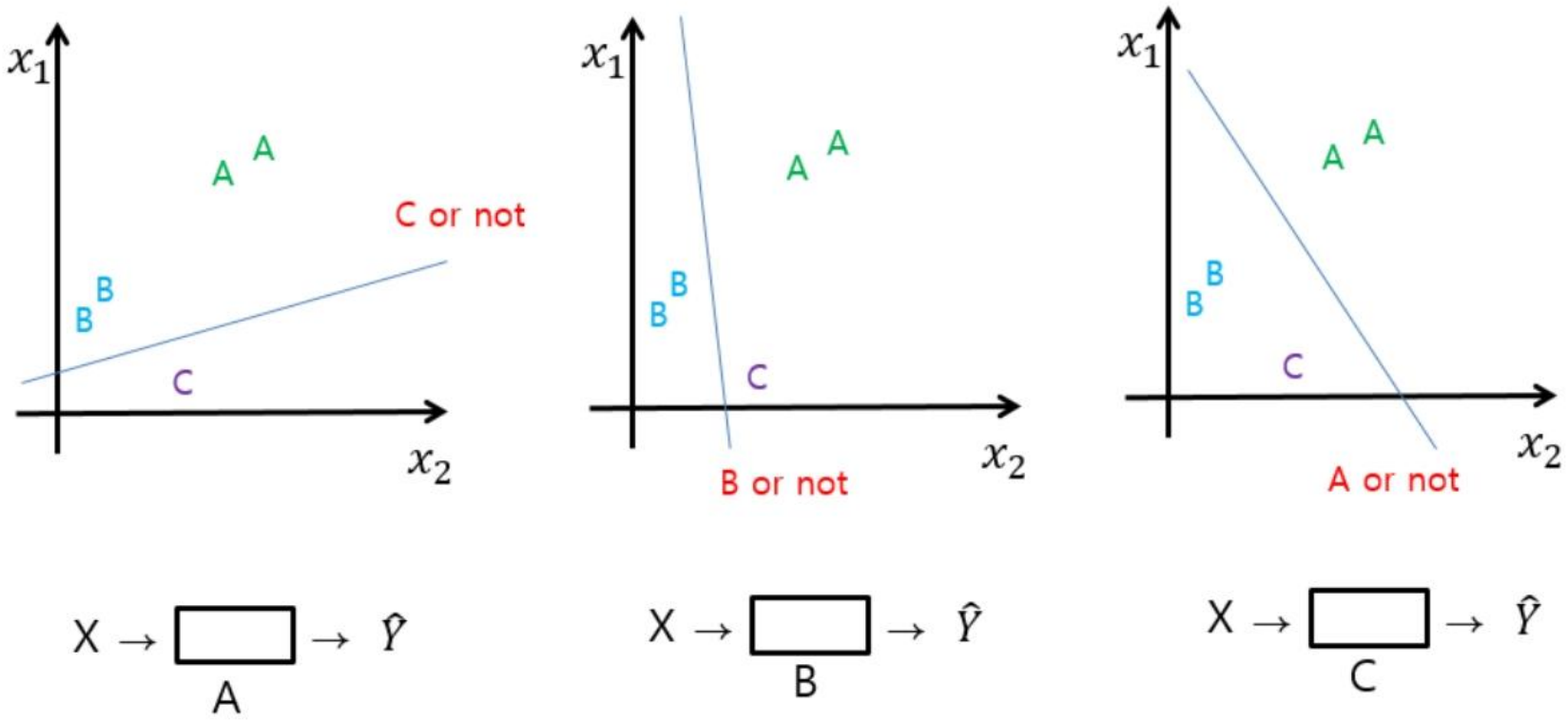


MULTINOMIAL CLASSIFICATION

x1 (hours)	x2 (attendance)	y (grade)
10	5	A
9	5	A
3	2	B
2	4	B
11	1	C

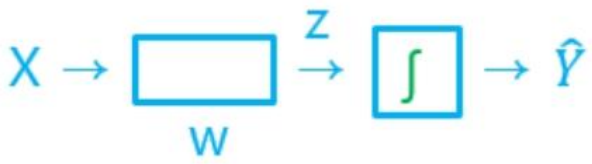


MULTINOMIAL CLASSIFICATION

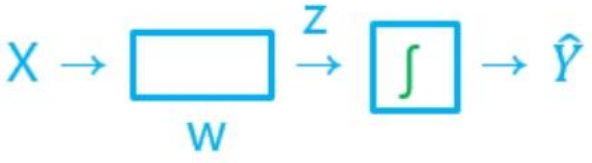


MULTINOMIAL CLASSIFICATION

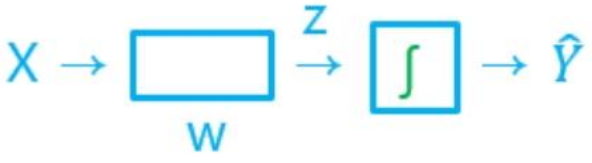
$$[w_1 \quad w_2 \quad w_3]\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = [w_1x_1 + w_2x_2 + w_3x_3]$$



$$[w_1 \quad w_2 \quad w_3]\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = [w_1x_1 + w_2x_2 + w_3x_3]$$



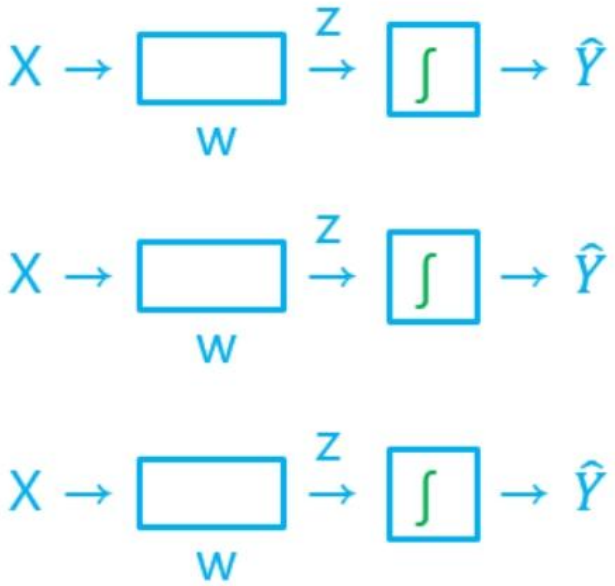
$$[w_1 \quad w_2 \quad w_3]\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = [w_1x_1 + w_2x_2 + w_3x_3]$$



MULTINOMIAL CLASSIFICATION

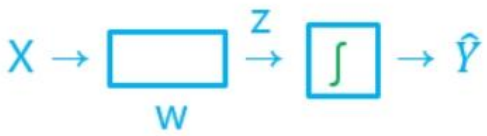
$$[w_1 \;\; w_2 \;\; w_3]\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = [w_1x_1 + w_2x_2 + w_3x_3]$$

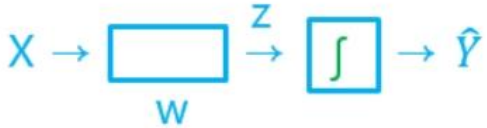
$$\begin{bmatrix} w_{A1} & w_{A2} & w_{A3} \\ w_{B1} & w_{B2} & w_{B3} \\ w_{C1} & w_{C2} & w_{C3} \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \;?$$

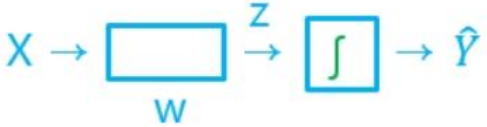


MULTINOMIAL CLASSIFICATION

$$\begin{bmatrix} w_{A1} & w_{A2} & w_{A3} \\ w_{B1} & w_{B2} & w_{B3} \\ w_{C1} & w_{C2} & w_{C3} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} w_{A1}x_1 + w_{A2}x_2 + w_{A3}x_3 \\ w_{B1}x_1 + w_{B2}x_2 + w_{B3}x_3 \\ w_{C1}x_1 + w_{C2}x_2 + w_{C3}x_3 \end{bmatrix} = \begin{bmatrix} \hat{Y}_A \\ \hat{Y}_B \\ \hat{Y}_C \end{bmatrix}$$



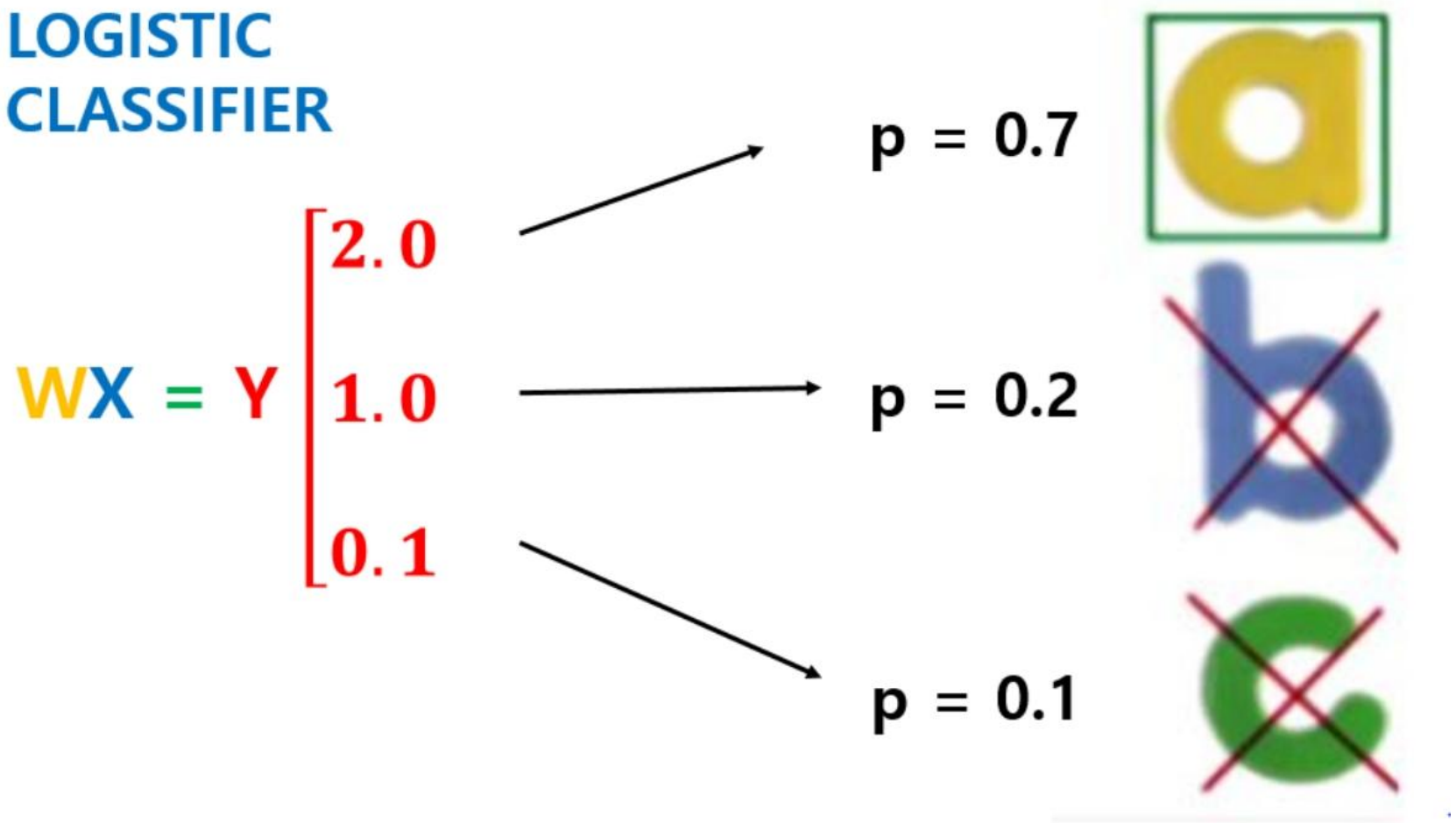




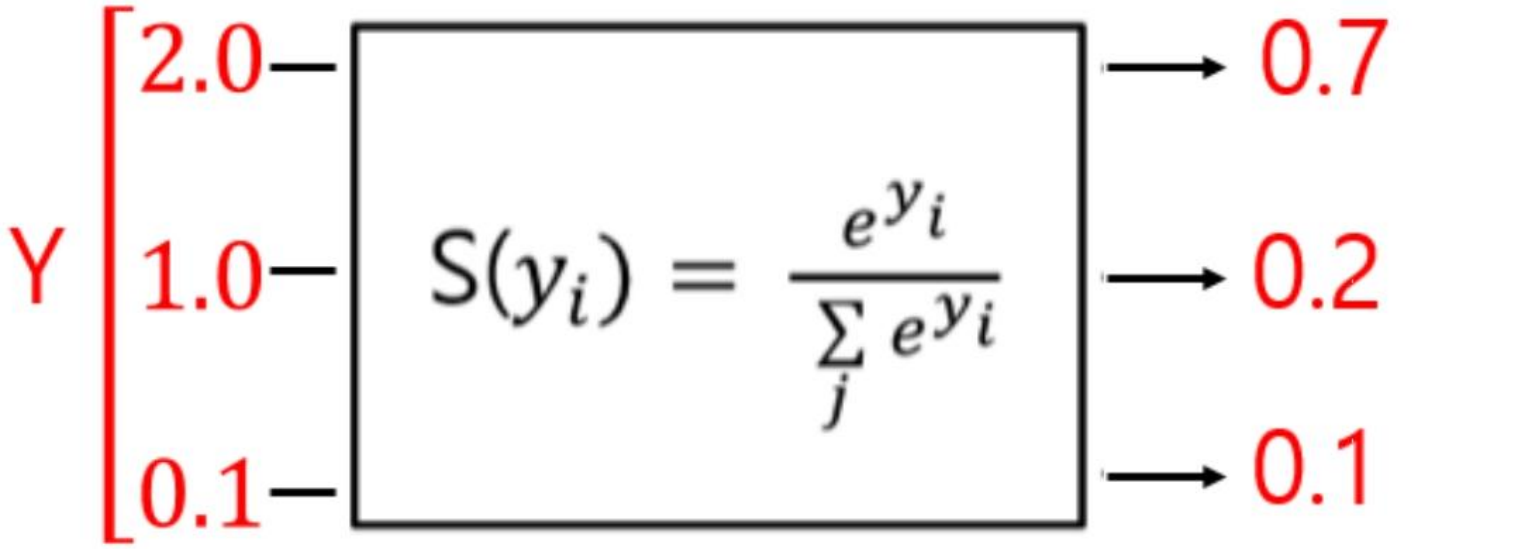
WHERE IS SIGMOID?

$$\begin{bmatrix} w_{A1} & w_{A2} & w_{A3} \\ w_{B1} & w_{B2} & w_{B3} \\ w_{C1} & w_{C2} & w_{C3} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} w_{A1}x_1 + w_{A2}x_2 + w_{A3}x_3 \\ w_{B1}x_1 + w_{B2}x_2 + w_{B3}x_3 \\ w_{C1}x_1 + w_{C2}x_2 + w_{C3}x_3 \end{bmatrix} = \begin{bmatrix} \hat{Y}_A \\ \hat{Y}_B \\ \hat{Y}_C \end{bmatrix} \begin{bmatrix} \mathbf{2.0} \\ \mathbf{1.0} \\ \mathbf{0.1} \end{bmatrix}$$

SIGMOID?



SOFTMAX



SCORES → PROBABILITIES

SOFTMAX

hypothesis = tf.nn.**softmax**(tf.matmul(X,W)+b)

XW = Y

tf.matmul(X,W)+b

$$\begin{bmatrix} 2.0 \\ 1.0 \\ 0.1 \end{bmatrix} \longrightarrow \boxed{S(y_i) = \frac{e^{y_i}}{\sum_j e^{y_i}}} \longrightarrow \begin{matrix} 0.7 \\ 0.2 \\ 0.1 \end{matrix}$$

SCORES ⟶ PROBABILITIES